



Book I. Prop. XIV. Theor. *If two straight lines (— and —), meeting a third straight line (—), at the same point, and at opposite sides of it, make with it adjacent angles (and) equal to two right angles; these straight lines lie in one continuous straight line. For, if possible let —,*

and not —, be the continuation of —, then $\text{yellow arc} + \text{blue arc} = \text{semicircle}$

but by the hypothesis $\text{yellow arc} + \text{blue arc} = \text{semicircle}$

$\therefore \text{blue arc} = \text{blue arc}$ (ax. III); which is absurd (ax. IX).

$\therefore$ — is not the continuation of —, and the like may be demonstrated of any other straight line except —, $\therefore$ — is the continuation of —.

Q. E. D.